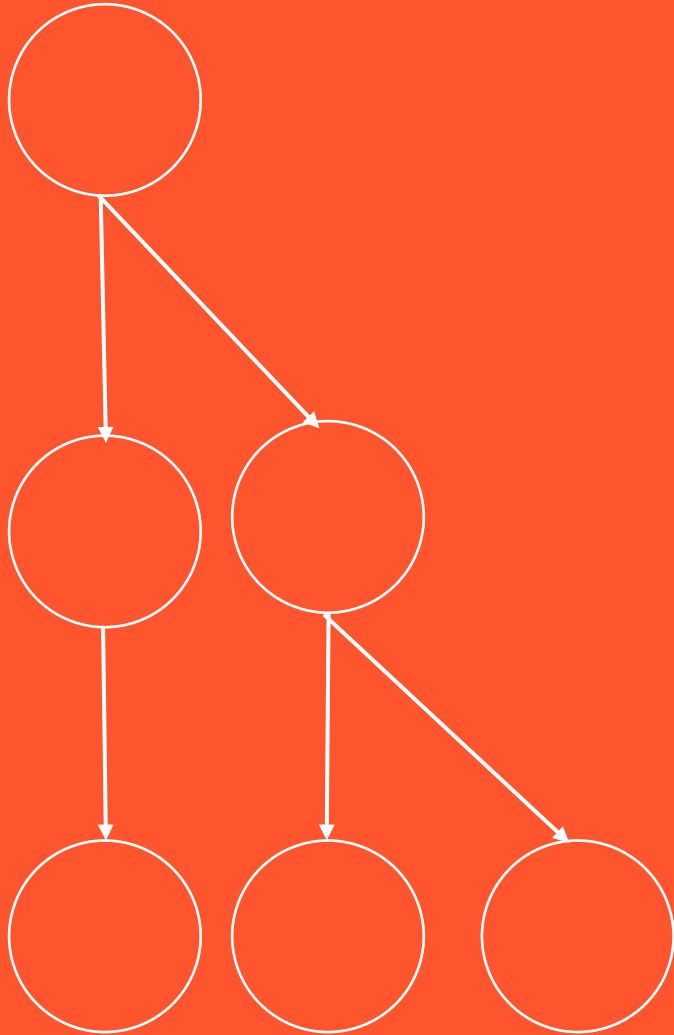


BCOG 100

Neural Circuits

Lecture 1: Neural Circuits in Brains



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 5-1

Password: circuit

Question: What's an example of a neural circuit described in the reading?

Introducing Neural Circuits

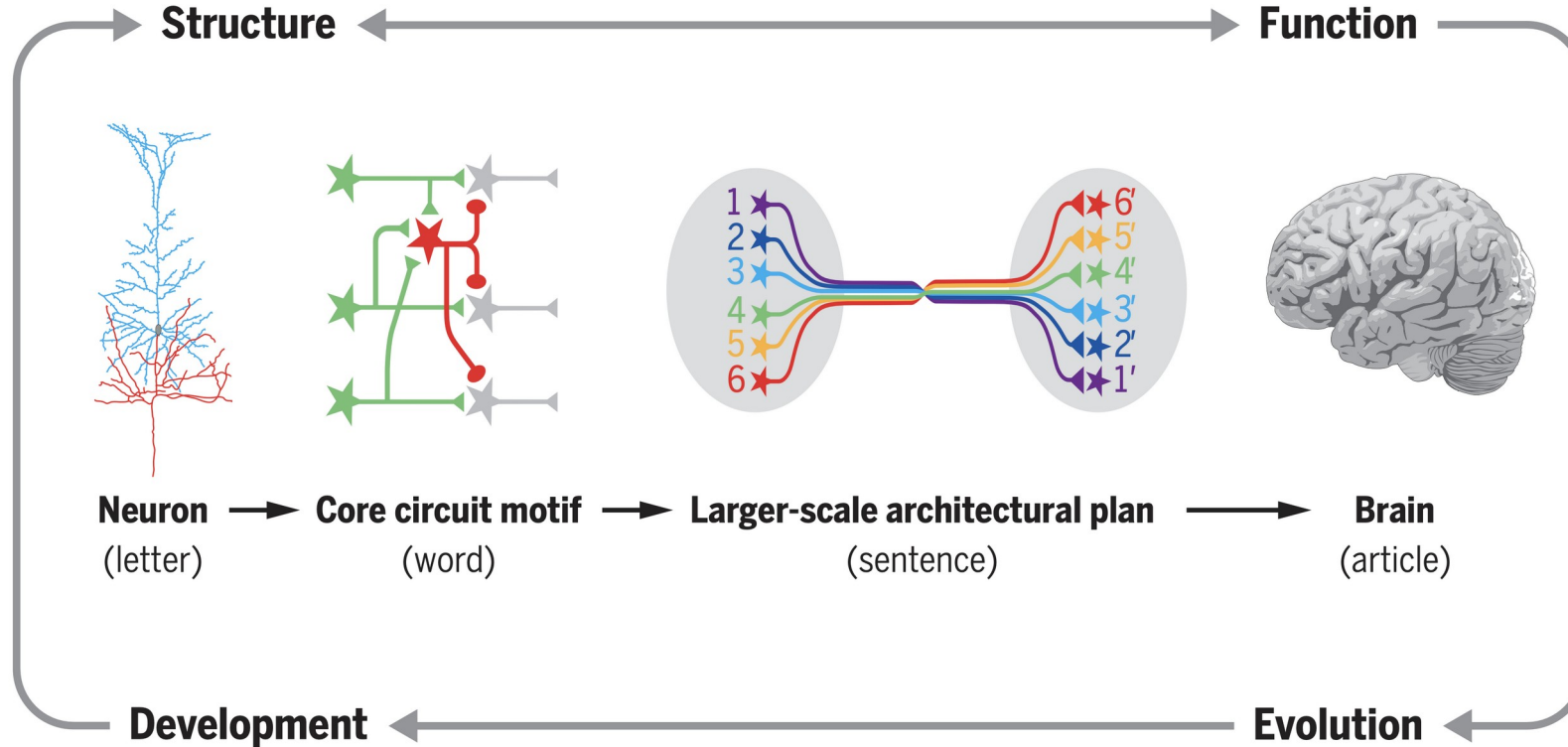
Recap & Overview

Last time: Evolution of neurons & brains

Today:

- How neurons work together in circuits
- Why circuits? From reflexes → computation

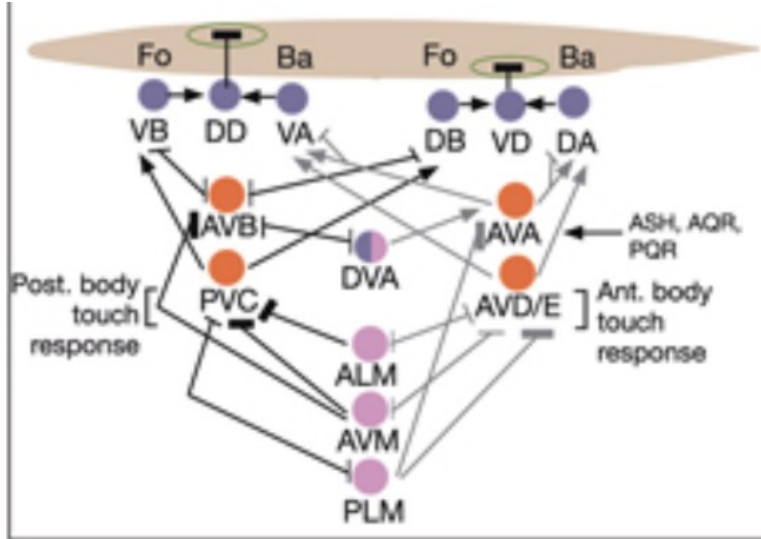
Introducing Neural Circuits



What is a Neural Circuit?

- Neurons don't act alone — they connect
- Circuits = patterns of connections
- “Letters → words → sentences” (Luo, 2021)

Introducing Neural Circuits



Mechanosensory neurons (*pink*) that detect touch stimuli synapse onto command interneurons (*red*). These, in turn, synapse onto motor neurons (*purple*) responsible for forward (Fo) and backward (Ba) locomotion, leading to rapid withdrawal from the stimulus. Spontaneous forward motion can be initiated by a head motor nervous system, which is partially independent of the command interneurons

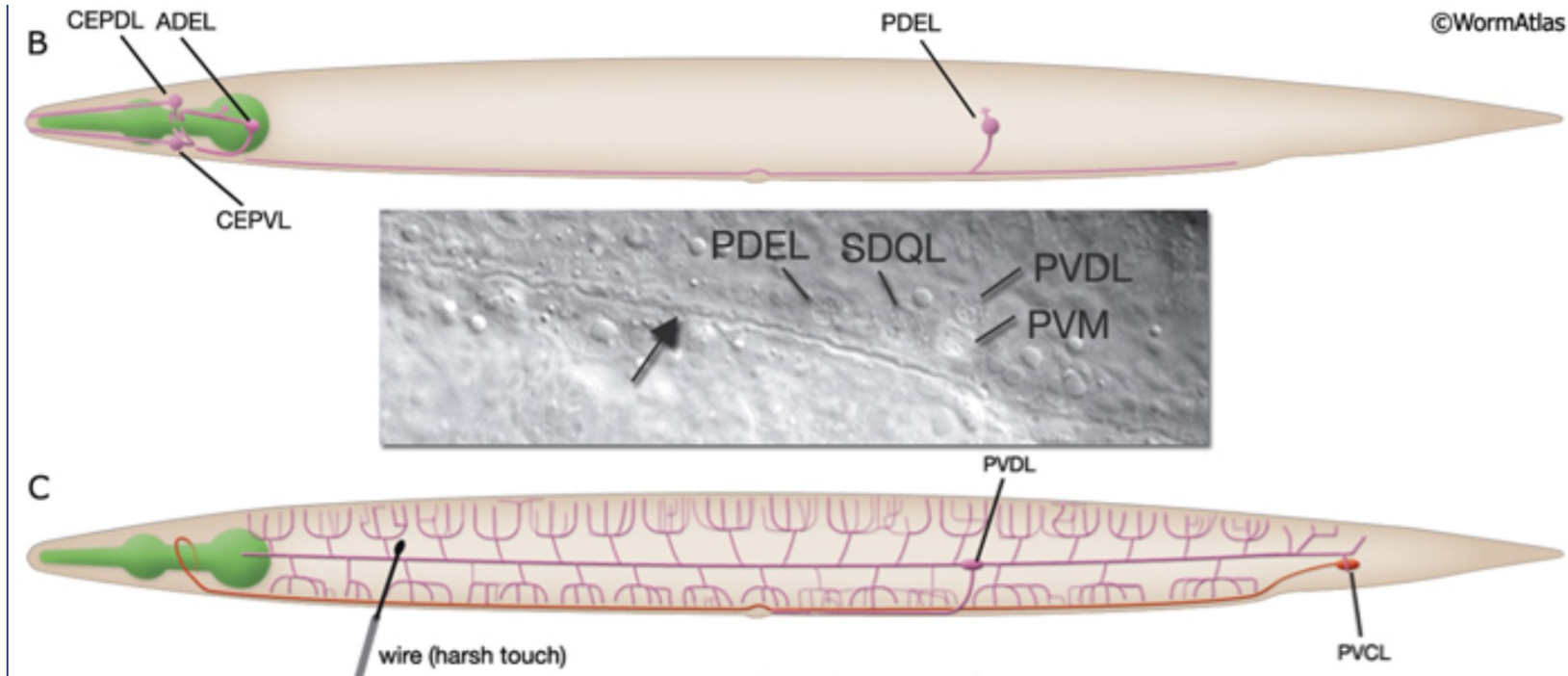
Simple Reflex Arc (*C. elegans*)

- Input → processing → output
- Example: touch reflex in *C. elegans*
- Circuits can be simple *and* powerful

Escape Circuits

- Circuits can trigger rapid survival behaviors
- Example: *C. elegans* escape response
- Example: human startle reflex

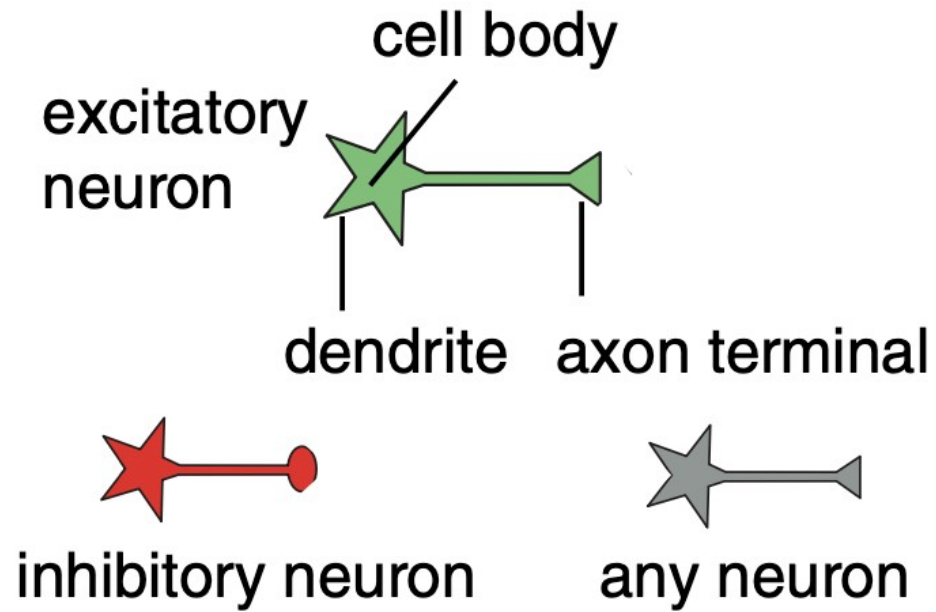
Introducing Neural Circuits



Sensory-Motor Integration Circuits

- Circuits combine inputs from multiple senses
- Interneurons integrate signals
- Output = coordinated behavior

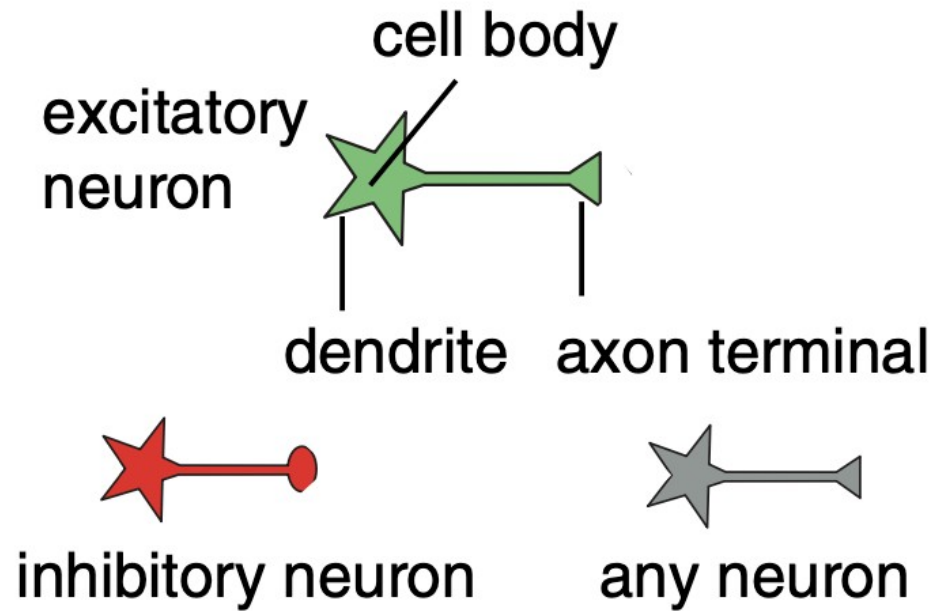
Types of Neural Circuits



Excitatory vs. Inhibitory Connections

- Excitatory: increase activity in target neuron
- Inhibitory: decrease activity in target neuron
- Circuits balance both to control output

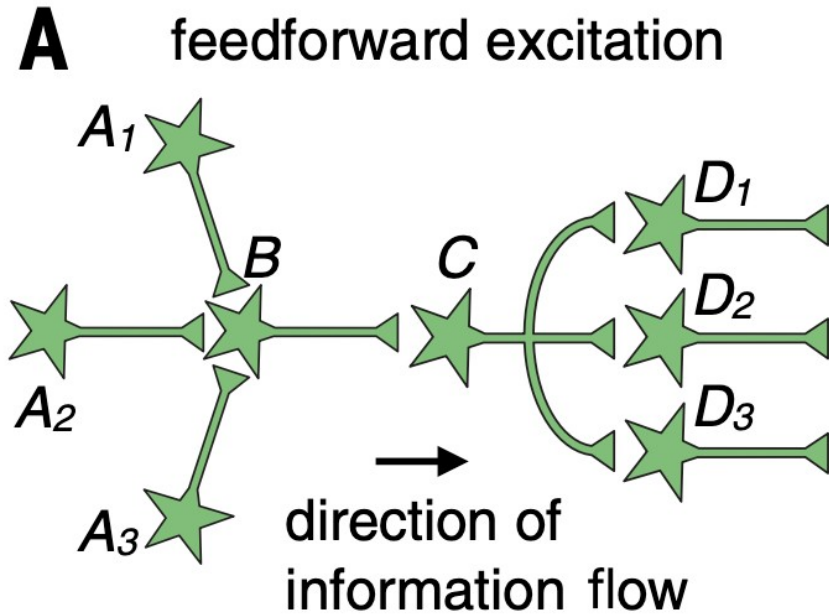
Types of Neural Circuits



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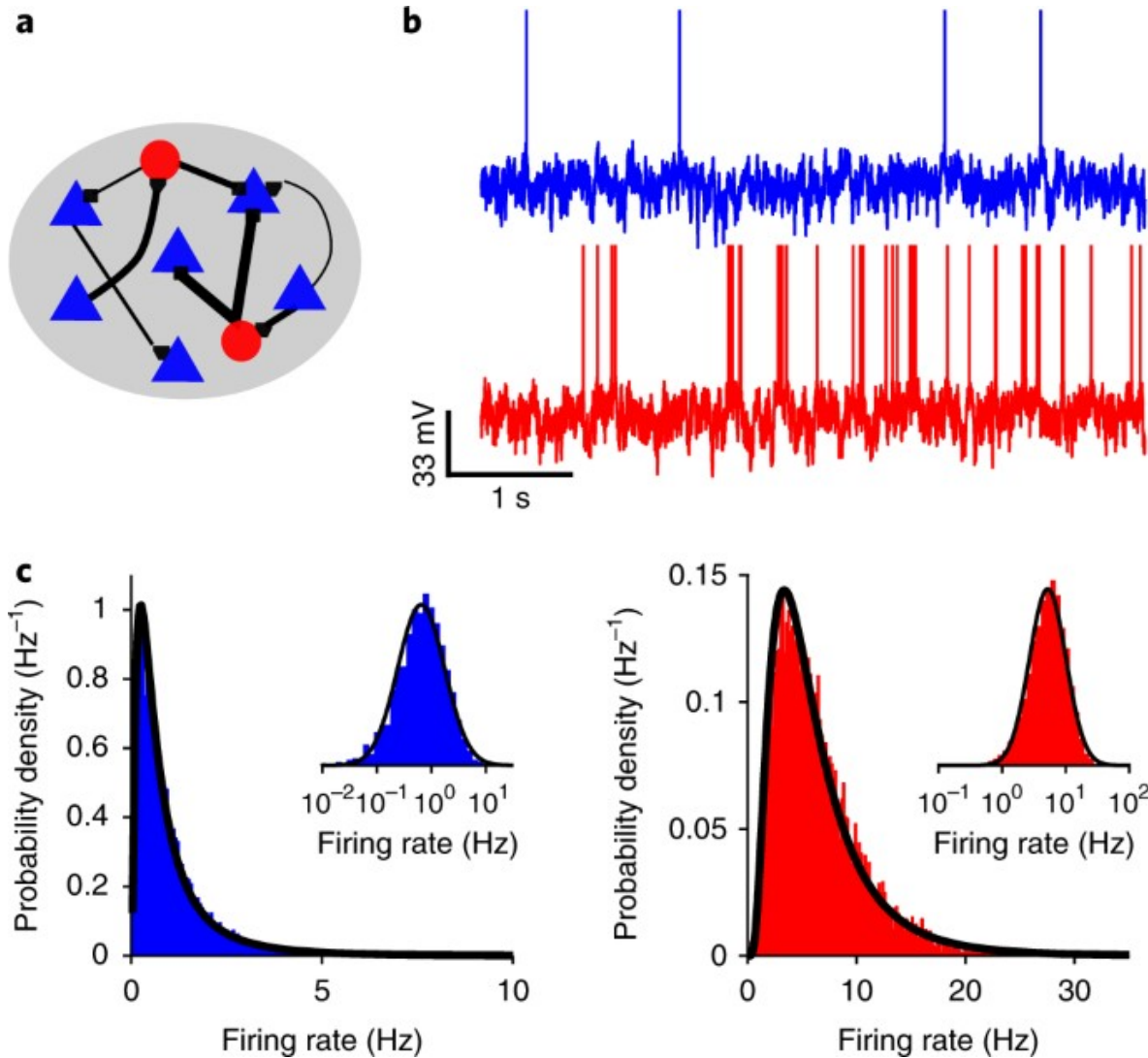
Types of Neural Circuits



Feedforward Circuits

- Feedforward circuits regulate timing, strength, stability
- Signals move in one direction
- Example: worm touch → motor response

Types of Neural Circuits



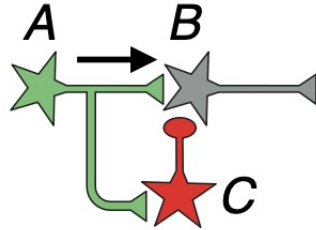
Feedback Circuits

- Feedback circuits regulate timing, strength, stability
- Output loops back to earlier stage
- Example: inhibition can gradually shut a circuit off

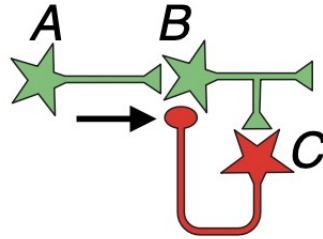
Types of Neural Circuits

B

feedforward inhibition

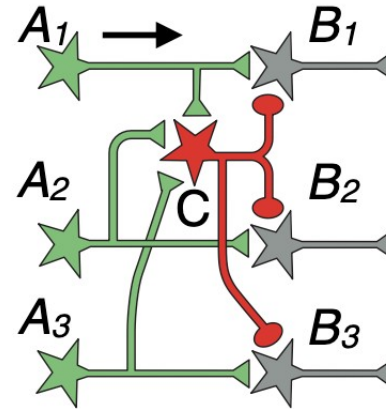


feedback inhibition



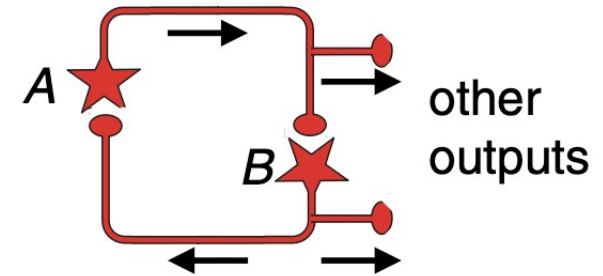
C

lateral inhibition



D

mutual inhibition

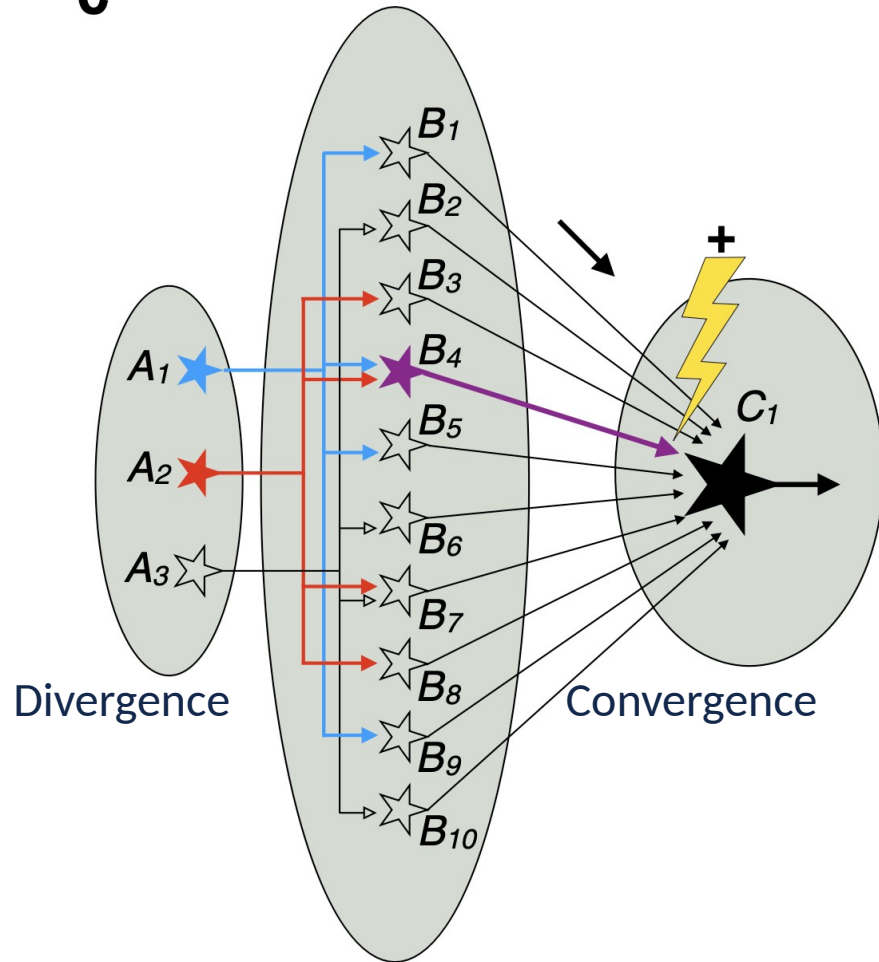


Feedback Circuits

- **Feedforward inhibition:** input excites inhibitor → sharpens response
- **Feedback inhibition:** output activates inhibitor → prevents runaway
- **Lateral inhibition:** neighbors inhibit each other → contrast, sharpening
- **Mutual inhibition:** two units inhibit each other → winner-takes-all

Types of Neural Circuits

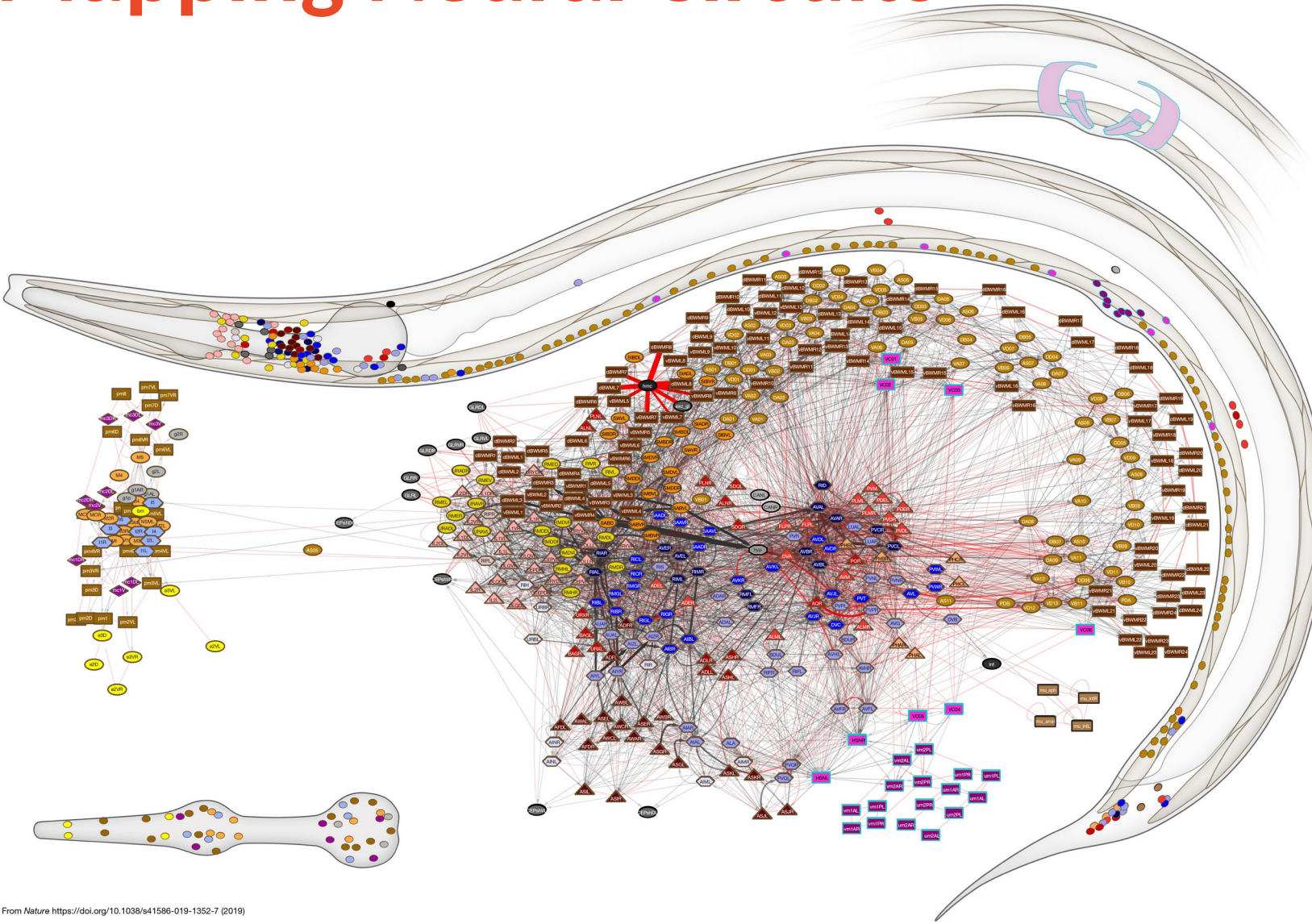
c



Divergence and Convergence of Signals

- **Convergence:** many inputs $\rightarrow$ one target
- **Divergence:** one input $\rightarrow$ many targets
- Enables integration & flexible output

Mapping Neural Circuits

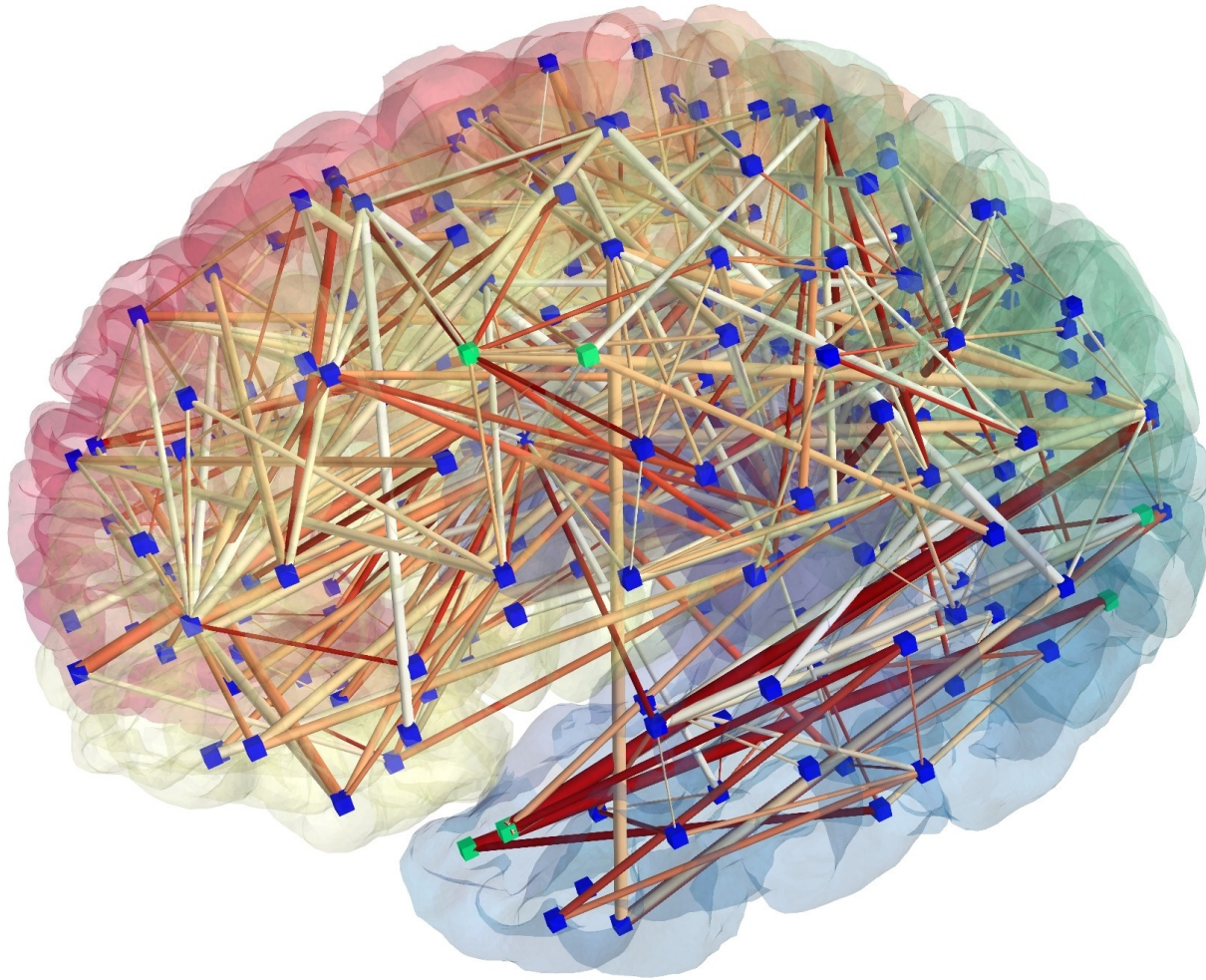


Connectomics Overview

- **Goal:** map who connects to who
- **Tools:** electron microscopy, tracing, imaging
- **Example:** *C. elegans* connectome (302 neurons)

From Nature <https://doi.org/10.1038/s41586-019-1352-7> (2019)

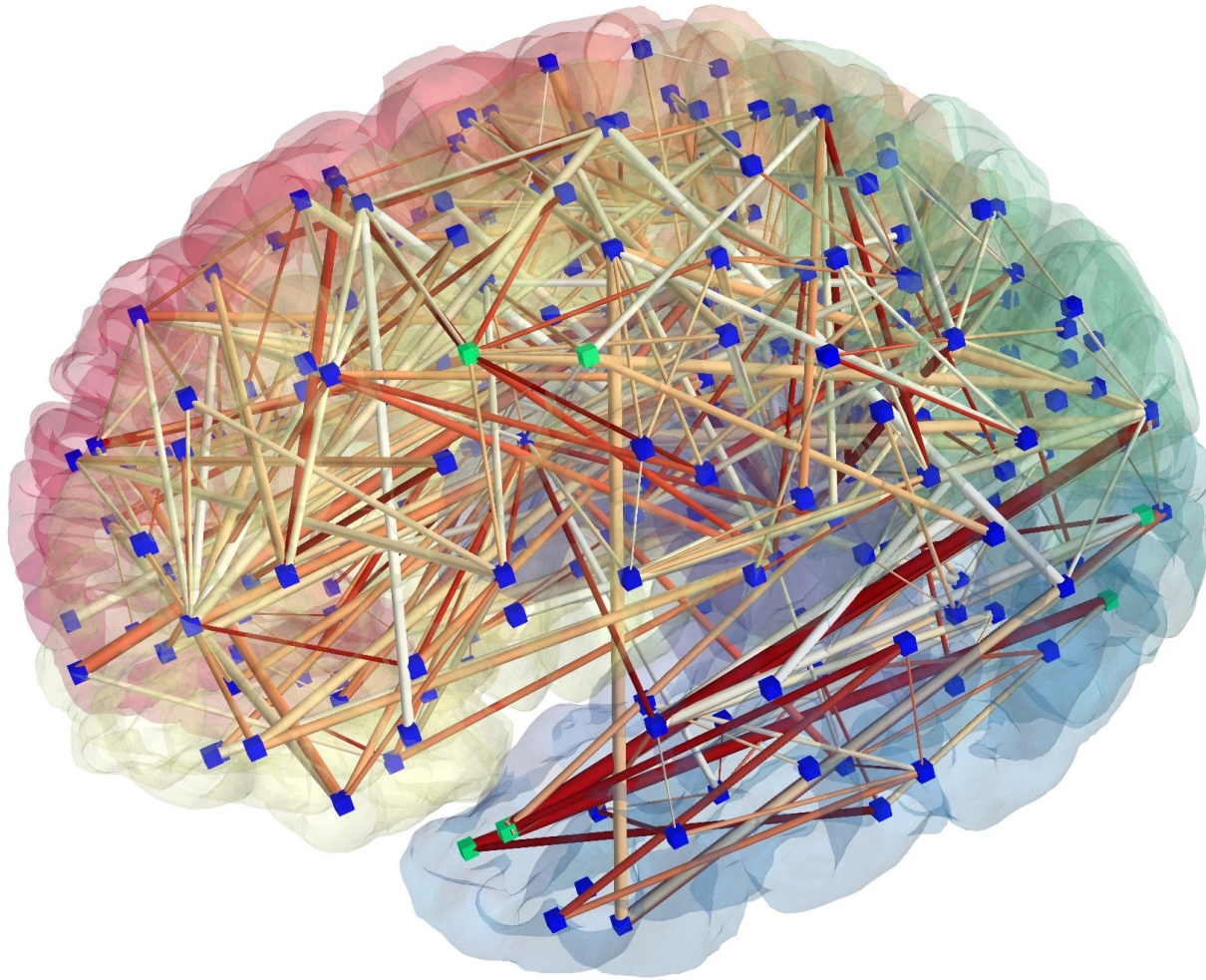
Mapping Neural Circuits



Human Connectomics

- Human brain: ~86 billion neurons
- fMRI & diffusion imaging map large-scale networks
- Examples: default mode network, language network,

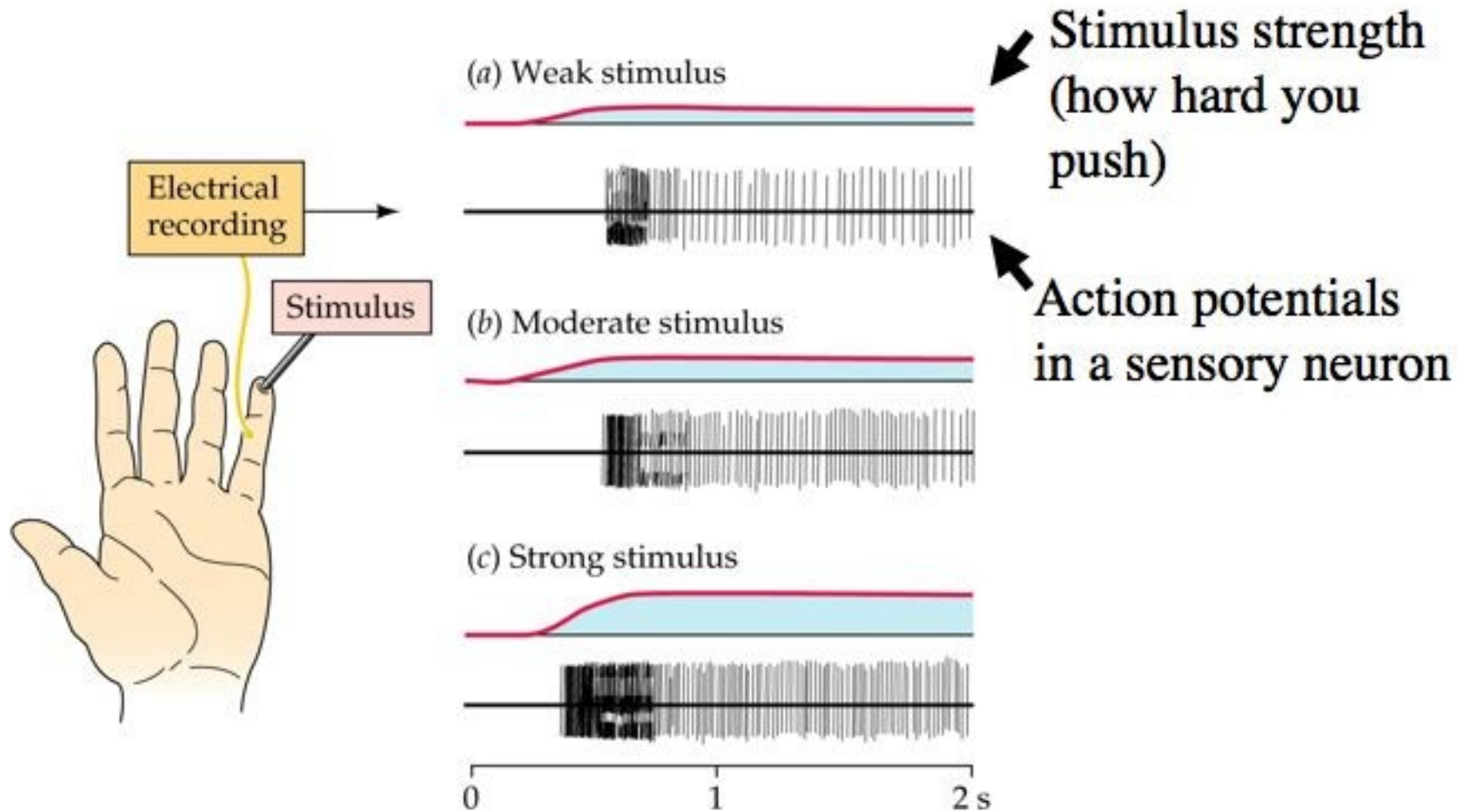
Mapping Neural Circuits



Example Human Brain Networks:

- Default Mode Network (DMN)
- Dorsal Attention Network (DAN)
- Ventral Attention Network (VAN)
- Salience Network (SN)
- Central Executive Network (CEN)
- Limbic Network
- Somatomotor Network
- Visual Network

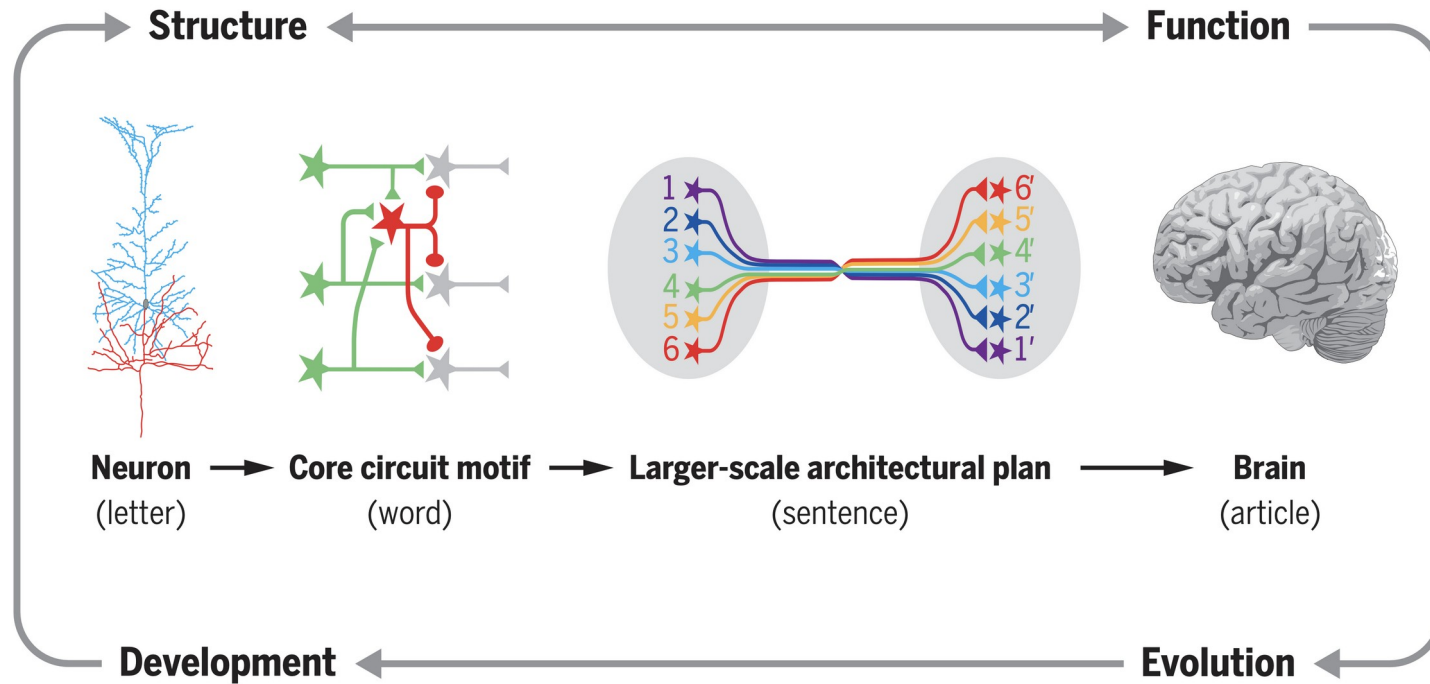
Mapping Neural Circuits



Neurons as Rate-Based Systems

- Real neurons signal by firing rates
- Spikes → average rate over time
- Not just “on” or “off”

Why Circuits Matter: From Neurons to Computation



Why Circuits Matter

- Circuits transform inputs into outputs
- Enable reflexes, rhythms, learning, thought
- Foundation for brain computation

Why Circuits Matter: From Neurons to Computation

Wrap-Up & What's Next

Today:

- what neural circuits are & why they matter
- Key motifs: excitation, inhibition, feedback, convergence
- Circuits = “words” of the brain

Next: how circuits adapt and learn from experience

